

## **Subject Description Form**

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| <b>Subject Code</b>                                                  | <b>ITC3227K</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Subject Title</b>                                                 | <b>Knitwear Technology II</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Credit Value</b>                                                  | <b>3</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Level</b>                                                         | <b>3</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Pre-requisite(s) /<br/>&lt;Co-requisite&gt; /<br/>(Exclusion)</b> | <ITC3222K Knitted Structure Design &<br><ITC3226K Knitwear Technology I >                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Objectives</b>                                                    | The subject provides a platform to learn through extensive experimentation in the areas of both knitted structure design and knitwear production.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Intended Learning Outcomes</b>                                    | <p>Upon completion of the subject, students will be able to:</p> <ol style="list-style-type: none"> <li>identify and describe the structure and characteristics of knitwear and the yarn used;</li> <li>describe, record and produce sample swatches comprising the essential and compound stitch variations applicable to the design and production of knitwear;</li> <li>demonstrate knowledge of the capability of the most sophisticated design software and automatic flat knitting machines essential to the production of ‘top end’ knitwear;</li> <li>communicate effectively with industrial practitioners;</li> <li>identify and apply the safety and ethical practices in the industry.</li> </ol>                                                                                         |
| <b>Subject Synopsis/<br/>Indicative Syllabus</b>                     | <p><b>(I) Yarns for Knitting</b><br/> Overview of yarn types suitable for knitwear including yarn construction and fiber contents.<br/> General effects of yarn characteristics on knit structure and knitted garments.</p> <p><b>(II) CAD/CAM Applications in Knitwear Design</b><br/> Application of appropriate CAD systems in knitwear design, including yarn design, stitch design, knitting programming and simulation.<br/> Application of up-to-date CAM knitting equipment for:<br/> Design and development of innovative knitted structures based on “package” method.<br/> Design and development of fully fashioned knitwear.</p> <p><b>(III) Seamless Knitting Techniques</b><br/> Introduction of seamless knitting principles.<br/> Demonstration of seamless knitwear production.</p> |

|                                                                        | <p>(IV) <b>Creative Knitted Structure Design Project</b><br/>Design creative knitted swatches with the combination of yarns and stitch development</p> <p>(V) <b>Safety and Ethical Practices in the Industry</b><br/>General safety and general ethical practices when operating the machines in knitwear industry</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                |   |   |   |          |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                              |             |   |   |   |   |   |                       |     |   |   |   |  |  |                         |     |   |   |   |   |   |                    |           |  |  |  |  |  |       |      |  |  |  |  |  |
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| <b>Teaching/Learning Methodology</b>                                   | <p>Studio sessions will be used with the audio-visual aids and in conjunction with the laboratory demonstrations.</p> <p>A design project will be assigned to reinforce the studied subject materials. Through design project presentation and peer critique, inadequate areas will be identified and improved.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                |   |   |   |          |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                              |             |   |   |   |   |   |                       |     |   |   |   |  |  |                         |     |   |   |   |   |   |                    |           |  |  |  |  |  |       |      |  |  |  |  |  |
| <b>Assessment Methods in Alignment with Intended Learning Outcomes</b> | <table border="1"> <thead> <tr> <th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr> <tr> <th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr> </thead> <tbody> <tr> <td><b>Continuous Assessment</b></td><td><b>100%</b></td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr> <tr> <td>1. <i>Assignments</i></td><td>40%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr> <tr> <td>2. <i>Final Project</i></td><td>60%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr> <tr> <td><b>Examination</b></td><td><b>0%</b></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>Total</td><td>100%</td><td></td><td></td><td></td><td></td><td></td></tr> </tbody> </table> <p>Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:</p> <p>Assignments will assess students' understanding of the advanced knitting techniques and complicated fabric structures based on electronic flat knitting technology. Individual project will be designed to allow students to learn and apply the machine knitting techniques along with the correct selection of yarns in developing stitches which express individual innovation and creativity. The swatches created in the final project will be further developed and utilized in knitwear portfolio and collection.</p> |                                                                                |   |   |   |          | Specific assessment methods/tasks | % weighting | Intended subject learning outcomes to be assessed (Please tick as appropriate) |  |  |  |  | a | b | c | d | e | <b>Continuous Assessment</b> | <b>100%</b> | ✓ | ✓ | ✓ | ✓ | ✓ | 1. <i>Assignments</i> | 40% | ✓ | ✓ | ✓ |  |  | 2. <i>Final Project</i> | 60% | ✓ | ✓ | ✓ | ✓ | ✓ | <b>Examination</b> | <b>0%</b> |  |  |  |  |  | Total | 100% |  |  |  |  |  |
| Specific assessment methods/tasks                                      | % weighting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Intended subject learning outcomes to be assessed (Please tick as appropriate) |   |   |   |          |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                              |             |   |   |   |   |   |                       |     |   |   |   |  |  |                         |     |   |   |   |   |   |                    |           |  |  |  |  |  |       |      |  |  |  |  |  |
|                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | a                                                                              | b | c | d | e        |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                              |             |   |   |   |   |   |                       |     |   |   |   |  |  |                         |     |   |   |   |   |   |                    |           |  |  |  |  |  |       |      |  |  |  |  |  |
| <b>Continuous Assessment</b>                                           | <b>100%</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ✓                                                                              | ✓ | ✓ | ✓ | ✓        |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                              |             |   |   |   |   |   |                       |     |   |   |   |  |  |                         |     |   |   |   |   |   |                    |           |  |  |  |  |  |       |      |  |  |  |  |  |
| 1. <i>Assignments</i>                                                  | 40%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | ✓                                                                              | ✓ | ✓ |   |          |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                              |             |   |   |   |   |   |                       |     |   |   |   |  |  |                         |     |   |   |   |   |   |                    |           |  |  |  |  |  |       |      |  |  |  |  |  |
| 2. <i>Final Project</i>                                                | 60%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | ✓                                                                              | ✓ | ✓ | ✓ | ✓        |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                              |             |   |   |   |   |   |                       |     |   |   |   |  |  |                         |     |   |   |   |   |   |                    |           |  |  |  |  |  |       |      |  |  |  |  |  |
| <b>Examination</b>                                                     | <b>0%</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                |   |   |   |          |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                              |             |   |   |   |   |   |                       |     |   |   |   |  |  |                         |     |   |   |   |   |   |                    |           |  |  |  |  |  |       |      |  |  |  |  |  |
| Total                                                                  | 100%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                |   |   |   |          |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                              |             |   |   |   |   |   |                       |     |   |   |   |  |  |                         |     |   |   |   |   |   |                    |           |  |  |  |  |  |       |      |  |  |  |  |  |
| <b>Student Study Effort Expected</b>                                   | Class contact:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                |   |   |   |          |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                              |             |   |   |   |   |   |                       |     |   |   |   |  |  |                         |     |   |   |   |   |   |                    |           |  |  |  |  |  |       |      |  |  |  |  |  |
|                                                                        | ▪ Studio                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                |   |   |   | 39 Hrs.  |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                              |             |   |   |   |   |   |                       |     |   |   |   |  |  |                         |     |   |   |   |   |   |                    |           |  |  |  |  |  |       |      |  |  |  |  |  |
|                                                                        | Other student study effort:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                |   |   |   |          |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                              |             |   |   |   |   |   |                       |     |   |   |   |  |  |                         |     |   |   |   |   |   |                    |           |  |  |  |  |  |       |      |  |  |  |  |  |
|                                                                        | ▪ Assignments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                |   |   |   | 35 Hrs.  |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                              |             |   |   |   |   |   |                       |     |   |   |   |  |  |                         |     |   |   |   |   |   |                    |           |  |  |  |  |  |       |      |  |  |  |  |  |
|                                                                        | ▪ Individual Project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                |   |   |   | 35 Hrs.  |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                              |             |   |   |   |   |   |                       |     |   |   |   |  |  |                         |     |   |   |   |   |   |                    |           |  |  |  |  |  |       |      |  |  |  |  |  |
|                                                                        | Total student study effort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                |   |   |   | 109 Hrs. |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                              |             |   |   |   |   |   |                       |     |   |   |   |  |  |                         |     |   |   |   |   |   |                    |           |  |  |  |  |  |       |      |  |  |  |  |  |

**Reading List and  
References**

**Recommended Textbooks**

Spencer, D. (2001), *Knitting technology a comprehensive handbook and practical guide*. 3<sup>rd</sup> edition, Woodhead publishing Ltd.

Ray, S. C. (2012), *Fundamentals and advances in knitting technology*. Woodhead Publishing India Pvt. Ltd, India.

Raz, S. (1993), *Flat knitting technology*. Universal Maschienfabrik, Westhausen.

Black, S. (2005), *Knitwear in Fashion*. Thames & Hudson, London.

Brackenbury, T. (1992), *Knitted Clothing Technology*. Blackwell Scientific Publications, Oxford, Boston.

**Electronic Resources**

Shimaseiki

[www.shimaseiki.jp](http://www.shimaseiki.jp)

Stoll

[www.stoll.de](http://www.stoll.de)

WGSN

[www.wgsn.com](http://www.wgsn.com)

Help file for computer design software (Shimaseiki and Stoll)